

Novel Hyperbolic Curves Unique to Prime Numbers

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Abstract

We present a novel property of prime numbers whereby each prime $p \in \mathbb{P}^+$ possesses a unique coordinate pair—denoted as *pFrequency* (*pFr*) and *pFactor* satisfying the equation $pFr \times pFa = p - 1$. (*pFr*) is the lowest index of a primes divisibility in a form of OEIS A002450. Each prime number is unique, so its associated value $p - 1$ is likewise distinct. Since all $p > 2$ (a special base case) are odd, it follows that $p - 1$ is even. Given numerous factors of $p - 1$, this seems mundane yet the exclusive pairing yields a ratio (*pRatio*) which links each prime to a single, discontinuous hyperbolic curve all having $R^2 = 1$. The value *pFa* maps primes into their of hyperbolic curves. The primary relationship, $p - 1 = pFa/pFr$, is reverse-engineered from a regression-derived expression: $pRatio = pFa^2/(p - 1)$. (*pFr*) is the index of a primes divisibility in a for of OEIS A002450.

1 Introduction

This study analyzes primes within the interval $[5, 100000)$, offering high-precision examples, methodology, and procedures for computing *pFr*, thereby determining the corresponding *pFa* and related hyperbolic trajectory. We begin with progressively magnified visualizations of *pRatios* across this domain.

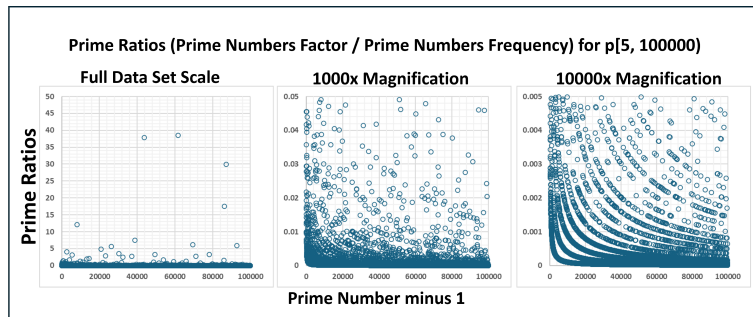


Figure 1: Initial plot of of primes $p[5, 100000)$ resembles stochastic noise. Upon 1000x magnification, patterns emerge. At 100000x, distinct curves become evident. The lowest curve on the left is the reciprocal function $y = x^{-1}$, here re-expressed as $y = pRatio = pFa^2/(p - 1)$, the special case where $p = 2$, $pFr = 1$, and $pFa = 1$, the singular case of an odd *pFr*.

2 Equations, Table Constructed A002450 vs. Contiguous A002450, Methodology, and Identification

The core functions originate from an auxiliary study on Mersenne primes. The method creates a list A002450 with alternating spaces, $[0, , 1, , 5, , 21, , 85, , 341, , \dots]$ ¹. This is not required but helps explain some properties.

2.1 Equations

1. $k = p - 1$
2. $f(k) = 2^k$
3. $g(x) = (f(k) - 1) \div 3$. Depicted in two rows.

2.2 Table Construction

1. Row 1: Construct a sequential row of $k \in \mathbb{N}_0$, start at $k = 0$. It tracks iterations, list indices, and relates to $p - 1$
2. Row 2: Compute $f(k) = 2^k$.
3. Row 3 Step 1, signifies $p = 2, pFr = 1, pFa = 1$: $g(x)_{Step1} = f(k) - 1$
4. Row 4 Step 2, signifies $p = 3, pFr = 2, pFa = 1$: $g(x)_{Step2} = g(x)_{Step1} \div 3$. A whole number created every second k -iteration, notable when k is even
5. Row 5 onward: For each prime $p \in [5, 100000)$, test for its divisibility within Row 4 to identify integer quotients.

Figure 2: Composite table for primes in the range $[5, 31]$

Observation: for prime p, beginning the k = p-1 iteration, p or p/q (mod g(f(2^k))) = 0 for some factor p-1/q (mod g(f(2^k)))																						
k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
f(x)	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576	
dR(g(f(x)) - Step 1 (-1))	0	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767	65535	131071	262143	524287	1048575	
- Step 2 (f/3) >>> g(f(x))		1		5		21		85		341		1365		5461		21845		87381		349525		
5				1				17					273				4369				69905	
7						3							195						12483			
11										31											31775	
13												105										
17								5									1285					
19																			4599			
23																						
29																						
31											11										11275	

2.3 Methodology and Observational Framework

- Periodic divisibility patterns arise (e.g., for $p = 5$, repetition occurs every $p - 1 = 4$ elements, at indices 4, 8), suggesting a definable frequency for the A002450-derived sequence that had to be even.²

¹Comparable to OEIS sequence A002450. An efficient recursive formulation is $a(n+1) = 4a(n) + 1$, with $a(0) = 0$ (see <https://oeis.org/A002450>)

²When the A002450 is contiguous, Row 1 (iteration index) may be optimized using $k_{next} = k + 2$ instead of $k + 1$ and the k th index will need to be multiplied by 2

- This periodicity implies a unique transformation but can occur at sub-multiples of $p - 1$; hence the need to check the list from the start (though there are faster methods).
- Plots of the derived pRatio expression ($pRatio = pFa^2/(p - 1)$) yielded the hyperbolic formations shown in Figure 1.

Figure 3: Fig. 2 with arrows showing periodicity where some have $pFr = p - 1$ and $pFa = 1$, e.g., $p = 5, 7, 11$ yield $pFr = 4, 6, 10$. However, pFr can be less than $p - 1$, e.g., $p = 31$ divides into 341 at the 10th index, so $pFr = 10$ and $pFa = (31 - 1)/10 = 3$.

Table 1 - Observation: for prime p, at iteration k = p-1, $g(f(2^k)) \equiv 0 \pmod{p}$ or $p \cdot q$, where $q \geq 3$

k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
f(x)	1	2	4	8	16	32	64	128	256	512	1024	2048	4096	8192	16384	32768	65536	131072	262144	524288	1048576
dR(g(f(x)) - Step 1 (-1))	0	1	3	7	15	31	63	127	255	511	1023	2047	4095	8191	16383	32767	65535	131071	262143	524287	1048575
dR(g(f(x)) - Step 2 (/3))	0	1	5	21	85	341	1365	5461	21845	87381	349525	69905	31775	12483	4999	1959	783	311	124	50	20
Prime : 5	0					1 5-1 cells							17 5-1 cells								
7	0						3 7-1 cells							273 5-1 cells				4369 5-1 cells			
11	0											31 11-1 cells									
13	0												105								
17	0													5 17-1/2 cells				1285 17-1/2 cells			
19	0																		4599		
23	0																				
29	0																				
31	0																				11275

3 Selected Examples

Figure 4: Power regression line illustrating the distribution of $p - 1, pRatios$ values for primes associated with $pFa = 19$.

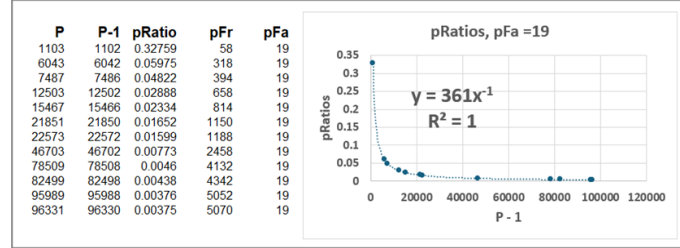
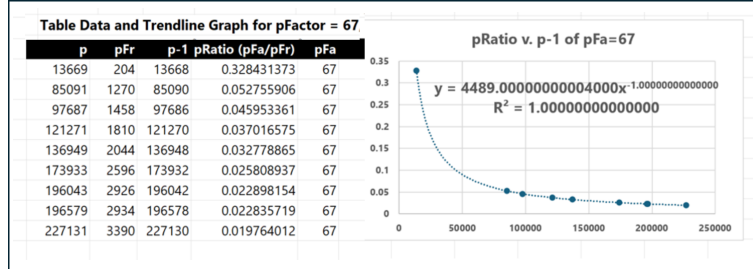


Figure 5: Like, Figure 4, $pFa = 67$ is plotted but with greater decimal accuracy.



4 Analysis and Observations

- The conditions under which a new pFactor curve, that is when pFa changes for the next prime remains indeterminate.
- The initial pFa yields the highest $pRatio$ due to its association with the smallest $p - 1$, given their inverse relationship.
- Prime (pFr, pFa) coordinate pairs are distinct across the entire domain.
- Higher-order pFactors do not necessarily emerge from larger prime values (see Table 1).
- The nature of triggering a new pFa curves remain elusive.

Figure 6: Four curves representing distinct pFa values, each with unique origins and no evident initiation pattern for subsequent curves. $y = pRatio = pFa^2/(p - 1)$ formulas are easy to identify.

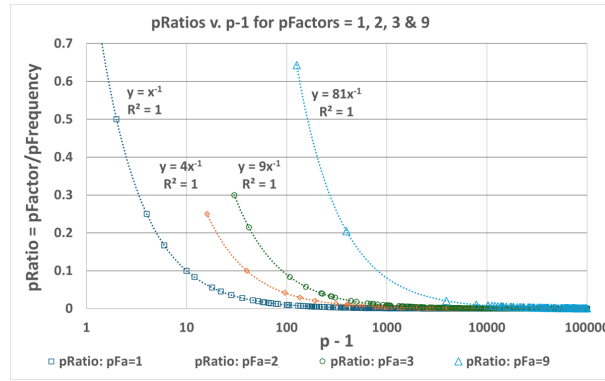


Figure 7: An earlier method testing decimal precision, involving additional variables beyond this paper's scope.

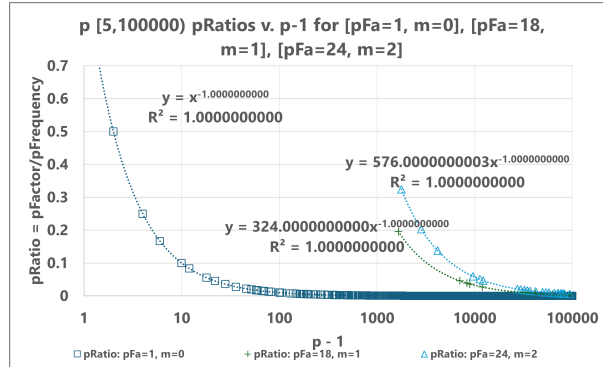


Figure 8: Clustered analysis of evenly spaced pFa curves (every 9th pFa). Subsequent curves exhibit decreased coordinate densities.

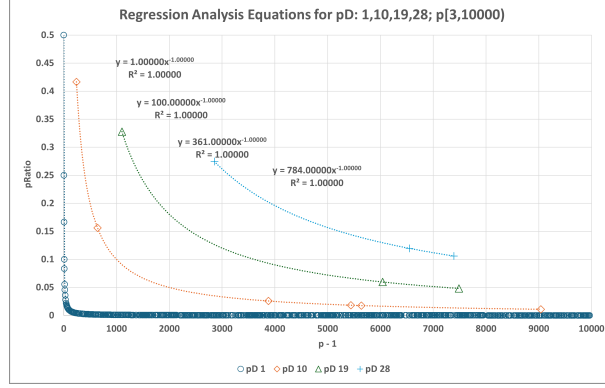


Table 1: First 30 pFactor Values and Their First Primes for p[5,100000)

pFa	1st Prime	pFa	1st Prime	pFa	1st Prime
1	5	11	331	21	2143
2	17	12	601	22	1321
3	31	13	4421	23	18539
4	73	14	673	24	1777
5	151	15	3061	25	2351
6	433	16	257	26	37441
7	631	17	1429	27	2971
8	337	18	1657	28	2857
9	127	19	1103	29	683
10	241	20	3121	30	17401

Figure 9: Histogram for primes in [5,100000). Bin [1,10] accounts for 92.2% (8841 of 9590) of pFactors.

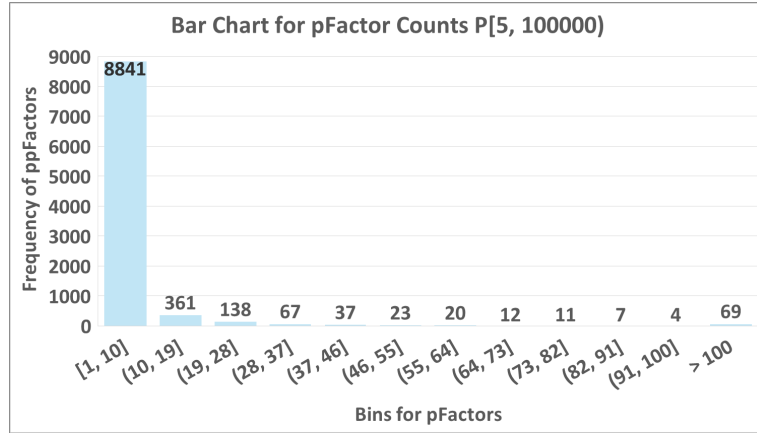


Table 2: Bin [1,10] Quartile Data, pFa 1 has 5541 of 9590 or 56.4% pFactors yet there's notable low coefficient of variance for pFa = 1 at 1.1% and within the bin at 0.6%.

pFactor	Q1	Q2	Q3	Q4	Mean	Std Dev	CoV	pFa Totals
1	1334	1363	1347	1367	1353	15.20	1.1%	5411
2	230	225	223	239	229	7.14	3.1%	917
3	252	252	226	231	240	13.72	5.7%	961
4	179	159	182	170	173	10.34	6.0%	690
5	78	49	75	43	61	17.82	29.1%	245
6	44	44	33	43	41	5.35	13.1%	164
7	35	28	35	43	35	6.13	17.4%	141
8	44	41	39	41	41	2.06	5.0%	165
9	21	31	25	31	27	4.90	18.1%	108
10	9	6	14	10	10	3.30	33.9%	39
QTotal	2226	2198	2199	2218	2210	13.96	0.6%	8841

5 Conclusions

Although the pFactors within the range [5, 100000) show considerable concentration within lower bins, this sample represents only a limited subset of the prime spectrum. While the coordinate structure ($pRatio, p - 1$) across successive hyperbolic curves lacks a recognizable pattern in initiating, unraveling its origin may provide deeper insights into the fundamental nature of prime numbers. Another compelling area for future exploration involves conditions under which a prime lies on a higher-order hyperbola rather than the most prevalent curve defined by $pRatio = pFa^2/(p - 1)$, with the base case $p = 2, pFr = 2, pFa = 1$.